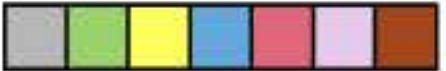


Input

Multi-Annotator
Dialog Corpus

Speaker A
I had a great time at
the concert!



Speaker B
Oh cool! I wish I
could've gone.



Speaker A
It started raining on
the way home.



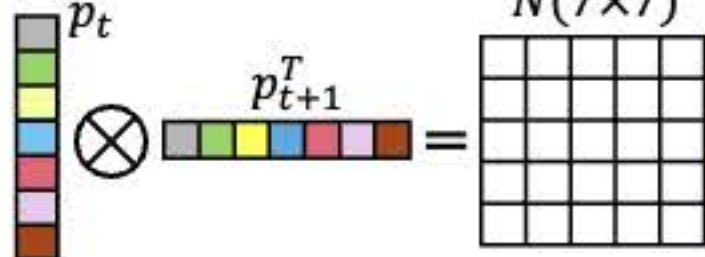
Speaker B
Ugh, that's the worst.



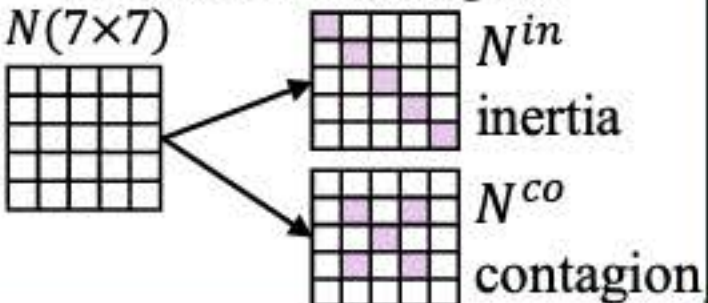
Stage 1

Bayesian Posterior

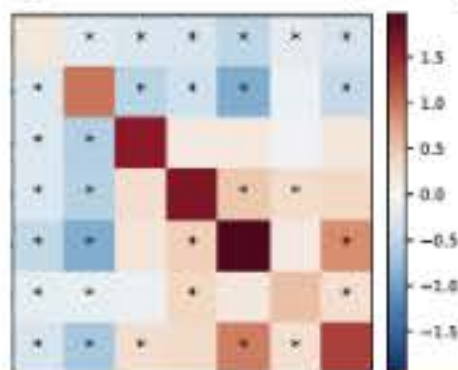
① Outer Product
→ Soft Transition Counts
 $N(7 \times 7)$



② Decompose Counts into
Inertia and Contagion
 $N(7 \times 7)$



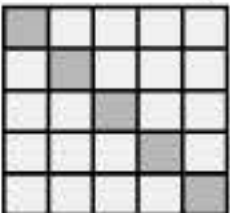
③ Posterior Mean of
Transition Matrix
 $\bar{T}$ posterior mean (7×7)



Stage 2

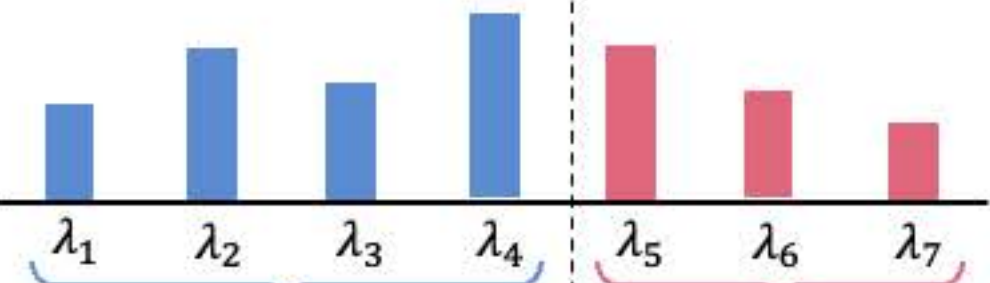
Spectral Decomposition

① Symmetrization $A(\text{symmetric}, 7 \times 7)$

$$A = \frac{\bar{T} + \bar{T}^T}{2}$$


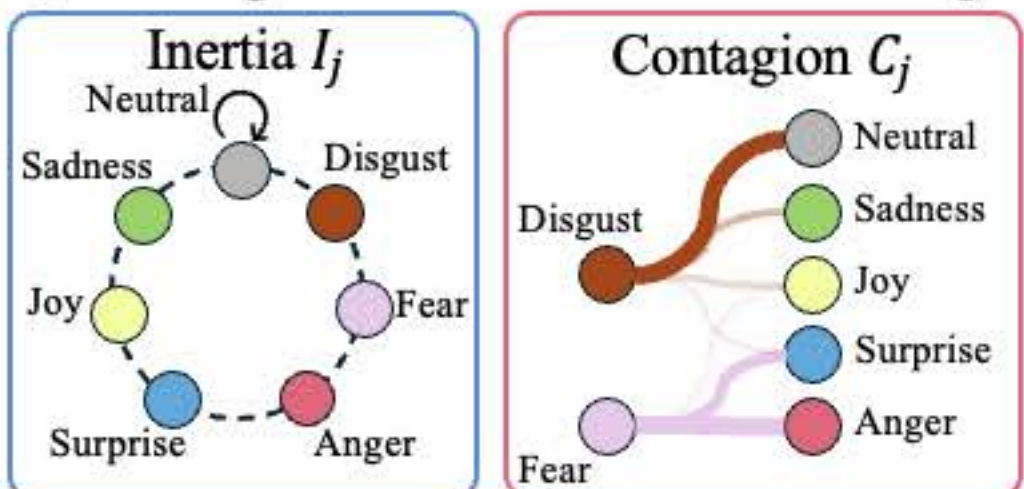
Symmetric matrix

② Graph Laplacian Eigendecomposition



Low-Frequency High-Frequency
Median λ^*

③ Decomposition into Inertia and Contagion



Output

Psychological Validation

- Plutchik Adjacency
- Gottman Cascade
- Hatfield Contagion